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Multi-omics approach characterizes the role of Bisphenol F in disrupting hepatic lipid metabolism

Yun Fan^{a,b,c,1}, Shiqi Li^{b,1}, Xiancheng Yang^{b,1}, Shengjun Bai^b, Min Tang^b, Xueer Zhang^b,
Chuncheng Lu^b, Chenbo Ji^a, Guizhen Du^{b,*}, Yufeng Qin^{a,b,c,*}

^a Women's Hospital of Nanjing Medical University (Nanjing Maternity and Child Health Care Hospital), Nanjing Medical University, Nanjing, China

^b Key Laboratory of Modern Toxicology of Ministry of Education, School of Public Health, Nanjing Medical University, Nanjing, China

^c Department of Microbiology and Infection, School of Public Health, Nanjing Medical University, Nanjing, China

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ABSTRACT

Bisphenol F (BPF), a substitute for bisphenol A (BPA), is ubiquitous existed in various environmental media. Exposure to BPF may promote non-alcoholic fatty liver disease (NAFLD), while the potential mechanism is still unknown. In current study, we used *in vitro* and *in vivo* model to evaluate its hepatotoxicity and molecular mechanism. Using multi-omics approach, we found that BPF exposure led to changes in hepatic transcriptome, metabolome and chromatin accessible regions that were enriched for binding sites of transcription factors in bZIP family. These alterations were enriched with pathways integral to the endoplasmic reticulum stress and NAFLD. These findings suggested that BPF exposure might reprogram the chromatin accessibility and enhancer landscape in the liver, with downstream effects on genes associated with endoplasmic reticulum stress and lipid metabolism, which relied on bZIP family transcription factors. Overall, our study describes comprehensive molecular alterations in hepatocytes after BPF exposure and provides new insights into the understanding of the hepatotoxicity of BPF.

1. Introduction

Bisphenols are widely used in industrial and daily consumer product (Ma et al., 2022). Bisphenol A (BPA) is predominately used bisphenols to produce food containers, epoxy resins, plastic tableware, polycarbonate plastics and baby bottles (Vancamp et al., 2023). Given the wide exposure and potential adverse effects, BPA has been banned and substituted with structurally similar analogues, such as bisphenol S (BPS) and bisphenol F (BPF) (Hu et al., 2023). Emerging BPF has been detected in the various environmental media, food and body fluids (Lucarini et al., 2023; Tchen et al., 2022; Wang et al., 2015), suggesting multiple routes of human exposure. Increasing evidence proved that chromosomal aberrations, cellular dysfunction, DNA damage and other adverse health effects were associated with BPF exposure (Cabaton et al., 2009). Furthermore, BPF exposure has been linked to reproductive disorders, neurodevelopmental diseases, liver oxidative damage and metabolic syndromes (Kim et al., 2022; Meng et al., 2019; Wang et al., 2021).

Liver is recognized as one of the main target organs for bisphenols and numerous recently studies have observed that BPF can induce hepatic metabolomic perturbations (Cano et al., 2021; Sun et al., 2023; Wang et al., 2022). What's more, our group conducted the computational adverse outcome pathway (cAOP) to evaluate hepatic toxicity of 20 BPs and found that BPF was ranked the highest score of hepatic toxicity when compared with other BPs. Previous study also indicated that BPF significantly elevated the levels of alanine aminotransferase (ALT) and aspartate aminotransferase (AST) in plasma or liver of male mice, suggesting BPF was a hepatotoxic chemical could disrupt the liver function (Sun et al., 2023). A high-content screening assay was applied *in vitro* to evaluate the hepatic lipid accumulation of 13 bisphenol analogs and observed that BPF was ranked 4th in the severity of the toxic effect of lipid accumulation (Liu et al., 2020). However, epidemiological and rodent studies have yielded conflicting results regarding the potential of BPF to cause nonalcoholic fatty liver disease (NAFLD) (Peng et al., 2023; Wang et al., 2021; Zhang et al., 2023). Moreover, the

* Corresponding authors at: Women's Hospital of Nanjing Medical University (Nanjing Maternity and Child Health Care Hospital), Nanjing Medical University, Nanjing, China (Yufeng Qin). Key Laboratory of Modern Toxicology of Ministry of Education, School of Public Health, Nanjing Medical University, Nanjing, China (Guizhen Du).

E-mail addresses: guizhendu@njmu.edu.cn (G. Du), qiny@njmu.edu.cn (Y. Qin).

¹ The first three authors contributed equally to this study and they should be regarded as joint first authors.

cytotoxicity effects and underlying molecular mechanisms contributing to the hepatotoxicity of BPF remain largely unclear.

NAFLD is becoming a significant risk factor for chronic liver disease with a global rising prevalence (Ferguson and Finck, 2021). Environmental risk factors, particularly environmental endocrine disruptors (EDCs), have been proved to be linked to lipid metabolism and the development of NAFLD in numerous studies have been conducted in recent years (Treviño and Katz, 2018). During hepatic steatosis, abnormal lipid buildup frequently coincides with insulin resistance and is associated with disrupted endoplasmic reticulum (ER) protein homeostasis with hepatocytes (Lebeaupin et al., 2018). To adapt to endoplasmic reticulum stress (ERS) and maintain protein folding homeostasis, the host will activate a compensatory signaling pathway termed the unfolded protein response (UPR) (Song and Malhi, 2019). Previous studies have proved that misfolding of proteins in the endoplasmic reticulum with ERS results in hepatic steatosis (Ashraf and Sheikh, 2015; Rutkowski et al., 2008). ERS-induced hepatic steatosis is multifactorial (e.g. hyperlipidaemia, medications, inflammation) and can occur during the process such as very low-density lipoprotein (VLDL) secretion, fatty acid (FA) oxidation and lipogenesis (Malhi and Kaufman, 2011; Song and Malhi, 2019). Besides, EDCs exposure has been reported to cause NAFLD and hepatic inflammation via ERS pathway (Sun et al., 2020). However, no data exist regarding the role of ERS in NAFLD after BPF exposure.

A variety of epigenetic modifications, including changes in chromatin accessibility, non-coding RNAs, histone modifications and DNA methylation, have been implicated in regulating transcriptome that dictate various biological processes. It was reported that exposure to bisphenols resulted in dynamic changes of chromatin state and histone modifications (D'Cruz et al., 2022; Jung et al., 2022; Li et al., 2023). The chromatin state is regulated by transcription factors (TFs) and chromatin modifiers cooperatively, dysregulation of this intricate balance result in a wide range of diseases (Ji et al., 2023). Extensive evidence has established that the activating transcription factor/cAMP-responsive element-binding (ATF/CREB) proteins could respond to ERS and fine-tuning systemic homeostasis in the liver (Cui et al., 2021).

The multi-omics strategy (transcriptomics, epigenomics and metabolomics) offers an integrated insights into hepatotoxicity caused by environmental factors (Deng et al., 2023; Mizuno et al., 2022; Peiseler et al., 2022). In this study, we applied a multi-omics approach, integrating ATAC-seq, CUT&Tag, metabolomics and RNA-seq, to systematically assess the hepatotoxic effects of BPF. We hypothesized that BPF reprogrammed the expressions of lipid metabolic genes and ERS contributing to NAFLD. This study will provide new insight into the BPF's hepatotoxicity, further expanding our understanding of the epigenetic mechanisms in BPF exposure related NAFLD.

2. Materials and methods

2.1. Animals and treatment

Animal experiments were conducted with the approval by ethics committee in this study. Eight-week-old C57BL6 male mice were kept at room temperature (18–22°C), relative humidity (30–50 %) and light controlled environment (light/dark cycle = 12 h: 12 h) for entire treatment. The BPF concentrations observed in environmental sewage sludge from the U.S. EPA targeted national sewage sludge survey span the range from < 1.79 to 242 ng/g (Yu et al., 2015b). The pharmacokinetic study of BPF showed that no-observed-adverse-effect level (NOAEL) was determined to be 2 mg/kg/day for male and 5 mg/kg/day for female (Lee et al., 2022). Previous study also reported that BPF at 0.2 mg/kg/day elicited the most pronounced effect on hepatic lipid accumulation and body weight gain in mice (Zhang et al., 2023). Therefore, we set the low dose (100 µg/kg bw/day) in this study was corresponding to the environmental level of BPF and the level that potentially induced the hepatic lipid accumulation successfully. The high dose (1000 µg/kg

bw/day) was corresponding to its order of magnitude of NOAEL. The mice were randomly assigned to receive corn oil (control group) or BPF (100 and 1000 µg/kg bw/day, Macklin, Shanghai) corn oil solutions for three months and each group contained eight mice. Each replicate represented one independent mouse sample. Body weight was measured weekly at a regular time. All mice samples were rapidly collected after sacrifice and stored at -80°C for further experiments.

2.2. Serum biochemical analyses

Serum triglycerides (TGs), nonesterified fatty acids, low-density lipoprotein (LDL) and high-density lipoprotein (HDL) levels were measured with blood automatic biochemical analyzer (Hitachi, Japan) (Fan et al., 2020). At least 7–8 samples were used in each group for the experiment.

2.3. Histology and staining

The liver tissues were embedded in paraffin after being fixed in 4 % paraformaldehyde solution overnight. Then paraffin-embedded sections were stained with hematoxylin-eosin (H&E) to evaluate the tissue morphology. The liver sections were analyzed using Panoramic Scanner (3DHISTECH, Hungary). At least 3 samples were used in each group for the experiment.

2.4. AML12 cell culture

The AML12 cell line (Wu et al., 1994) is constructed from hepatocytes transgenic mice carrying human TGF-α gene (CD1 strain, line MT42) obtained from the American Type Culture Collection (ATCC, United States), and it belongs to primary immortalized cells and can be used to study the occurrence and development of liver diseases such as lipid metabolism, steatosis, and nonalcoholic fatty liver disease. The cells were cultured at 37 °C and 5 % CO₂ in DMEM/F12 containing 10 % fetal bovine serum (Gibco, USA), 1 % penicillin-streptomycin (Beyotime, China), 40 ng/mL dexamethasone, 5.5 µg/mL transferrin, 5 ng/mL selenium and 10 µg/mL insulin (YUCHI, China). After exposed to BPF (100 µM, Macklin, Shanghai) for 24 h, AML12 cells were collected for the following experiments. The control group was treated with 0.1 % DMSO (Sigma-Aldrich, United States) *in vitro*.

2.5. Cell viability assay

Cell viability assay was preformed to determine the BPF dosage. Briefly, AML12 cells were seeded in 96-well plates with a density of 5×10^3 cells per well and 5 replicates per group were established. Then cells were exposed to BPF for 24 h from 0.1 to 100 µM. After incubation for 24 h, the CCK-8 solution (Vazyme, China) in culture medium were added into the plate, incubated for another 1 h and then measured the absorbance to assess the cell viability at 450 nm.

2.6. Nile red staining

Cellular lipid droplets were stained by 1 µM Nile Red (Solarbio, China) and nuclei were stained with DAPI (Invitrogen, USA). The fluorescence intensity was quantified and photographed via fluorescent microscope (Zeiss, Germany). At least 3 replicates per group for this experiment.

2.7. RNA-seq and data analysis

RNA sequencing was carried out by Novogene Company (Novogene, China) using the Illumina Novaseq platform. Three samples per group (control group and 1000 µg/kg bw/day BPF group) were performed in hepatic RNA-seq. Clean reads were mapped to mm10 and gene expression levels were estimated using DESeq2 package v1.20.0 (Love et al.,

2014). Differentially expressed genes (DEGs) were defined with an $FDR < 0.05$, $|\log_2 \text{FoldChange}| \geq 1$. The Gene Ontology (GO) analysis was calculated by the ClusterProfiler package v4.2.2 (Yu et al., 2012).

2.8. Real-time quantitative PCR (RT-qPCR)

We utilized the RT-qPCR to evaluate the expression levels of lipid metabolism related genes involved in *de novo* lipogenesis or beta oxidation of fatty acids. Total RNA was extracted and then reversed to cDNA using HiScript III cDNA Synthesis Kit (Vazyme, China). The quantitative PCR was conducted using SYBR Green Master Mix Kit (Vazyme, China) on a LightCycler 480 System (Roche, Basel Switzerland). Three samples per group for this experiment. All gene primers were listed in the Supplementary Table 1.

2.9. Metabolomics analysis

Briefly, liver samples (50 mg) were fragmented and sonicated for 5 min on ice, then centrifuged for 15 min at 4 °C (11,000 rpm). About 0.5 mL supernatant was prepared for LC-MS/MS non-target metabolome detection analyses (BioTree, Shanghai) by using an UHPLC system (Vanquish, Thermo Fisher Scientific), with a UPLC BEH Amide column (2.1 mm × 100 mm, 1.7 μm) coupled to Q Exactive HFX mass spectrometer (Orbitrap MS, Thermo) (Fan et al., 2023; Fan et al., 2020). The raw data included 3 quality control (QC) samples and 16 experimental samples (BPF:1000 μg/kg bw/day, 8 samples per group). Then we conducted original data management, RSD ≤ 30 % was set to guarantee data quality (QA). Metabolite identification was according to the retention time combined with the accurate mass of commercial standards (Supplementary Table 2). Principle component analysis (PCA) was conducted to visualize the distribution of the samples. 95 % confidence interval (CI) was utilized as the threshold in PCA score plot to identify outliers in the dataset. Partial Least Squares Discriminant Analysis (PLS-DA) was also helped to depict group separation and distinguish substantially differentially expressed metabolites (DEMs). The KEGG enrichment analysis was performed by MetaboAnalyst (Version 5.0) (Pang et al., 2022).

2.10. Joint pathway analysis

The identified DEGs and DEMs were further used to identify key genetic and metabolic pathways by MetaboAnalyst (Version 5.0). We used raw *P*-values < 0.05 to pick out significant gene-metabolic pathway (Pang et al., 2022).

2.11. ATAC-seq, CUT&Tag and data analysis

The Omni-ATAC protocol was used to generate ATAC-seq libraries (Corces et al., 2017). Two samples per group (control and 1000 μg/kg bw/day BPF group or 100 μM BPF group) were performed in ATAC-seq or CUT&Tag experiments. Briefly, pelleted cells (50 K cells) were lysed in cold ATAC-Resuspension Buffer (RSB) including 0.01 % digitonin, 0.1 % Tween 20 and 0.1 % NP40. Nuclei were then resuspended in tagmentation mix consisting of 25 μL 2 × TD buffer, 2.5 μL transposase, 16.5 μL D-PBS, 0.5 μL 1 % digitonin, 0.5 μL 10 % Tween-20 and 5 μL H₂O. The reaction was incubated in a thermomixer for 30 min at 37 °C. DNA fragments were amplified for 9 to 12 cycles and purified by AMPure XP beads. DNA libraries were sequenced on Illumina NovaSeq platform.

H3K27ac CUT&Tag was conducted as previously described (Kaya-Okur et al., 2019). Briefly, 100 K cells were resuspended in lysis buffer containing 0.5 mM spermidine, 0.1 % Triton-X100, 1 × protease inhibitor, 10 mM KCl, 20 % glycerol and 20 mM HEPES on ice for 10 min. Then nuclei were fixed with 0.01 % formaldehyde for 2 min at room temperature and quenched using glycine (75 mM, final concentration). Next, we transferred nuclei into wash buffer (1 × protease inhibitor, 20

mM HEPES, 150 mM NaCl and 0.5 mM spermidine) and subsequently blended with concanavalin A-coated magnetic beads (ConA beads) with binding buffer (1 mM CaCl₂, 10 mM KCl, 1 mM MnCl₂ and 20 mM HEPES). The mixture was incubated with 0.5 μL H3K27ac antibody (ab4729, Abcam) at 4 °C overnight in 50 μL antibody buffer (2 mM EDTA and 0.1 % (wt/vol) BSA). After washing, the nuclei were mixed with home-made pG-Tn5 in dig-300 buffer (0.01 % digitonin, 300 mM NaCl, 20 mM HEPES, 0.5 mM spermidine and 1 × protease inhibitors) at room temperature for 1 h. The cells were resuspended in 300 μL fragmentation buffer containing dig-300 buffer with 10 mM MgCl₂ and incubated 37 °C for 1 h. Each sample was incubated for 1 h at 50 °C to stop fragmentation and solubilize DNA fragments. Finally, DNA was extracted and amplified to make H3K27ac CUT&Tag libraries.

Raw reads of ATAC-seq and CUT&Tag were filtered using fastp v0.23.1 software (Chen et al., 2018) and mapped to mm10 using bowtie2 v2.3.5 (Langmead and Salzberg, 2012). For unique alignments, deduplication was done by Picard MarkDuplicates v2.18.16 (<https://broadinstitute.github.io/picard/>, 2023/08/25). ATAC-seq and CUT&Tag peak calling were performed with MACS3 v3.0.0b1 with following parameters: -f BAMPE -B -q 0.01 -extsize 200 for sharp peaks; -f BAMPE -B -q 0.01 -nomodel -shift -100 -extsize 200 for ATAC peaks. Differential peaks and motif analysis were performed using HOMER software. ChipSeeker v1.36.0 package was used for peak annotation (Yu et al., 2015a). Heatmaps and profiles for normalized CUT&Tag and ATAC-seq were made by bamCoverage function of deepTools (Ramírez et al., 2014).

2.12. Statistical analysis

The means ± standard deviation (SD) were displayed for all data. GraphPad Prism 9.0 was used to create graphs and conduct statistical analysis. Two-tailed unpaired Student's *t* test or Mann-Whitney test was applied to compare differences between two groups. One-way analysis of variance (ANOVA) was used to test the difference among multiple groups. *P* < 0.05 or *FDR* < 0.05 was considered statistically significant as described in the text.

3. Results

3.1. BPF exposure disrupted hepatic lipid metabolism in vivo and in vitro

To investigate whether BPF exposure disrupt hepatic lipid metabolism, 8-week mice were treated with BPF at multiple doses (0, 100 and 1000 μg/kg bw/day) continuously for 3 months. Exposure to BPF (100 and 1000 μg/kg bw/day) significantly increased body weight (Fig. 1A). High dose of BPF exposure exhibited significantly higher liver weight while no statistical differences in white adipose tissue (WAT) hypertrophy (Fig. 1B-C). Compared with the control group, the level of serum LDL-C exhibited a dose-dependent increase in response to BPF exposure (Fig. 1D), while no statistical differences were observed in serum HDL-C and TG levels (Fig. 1E-F). Furthermore, 100 μg/kg bw/day BPF exposure elevated serum non-esterified fatty acids (NEFA) (Fig. 1G). We used H&E staining to further evaluate hepatocellular morphology and lipid deposition, no significant changes in hepatocyte steatosis and lipid droplets were observed in the two BPF-exposed groups (Fig. 1H).

To further evaluate the hepatotoxicity of BPF exposure, we used AML12 cell line as an *in vitro* model and selected 100 μM as exposure dosage basing on cell proliferation results (Fig. S1). Nile red staining results indicated that BPF exposure caused hepatic lipid deposition (Fig. 1I). Additionally, lipid metabolism related genes involved in *de novo* lipogenesis or beta oxidation of fatty acids, including *Acot13*, *Cpt1a*, *Acox2*, *Acox2*, were up-regulated, while *Elovl6* was down-regulated (Fig. 1J). Collectively, these results imply that BPF exposure disrupts hepatic lipid metabolism.

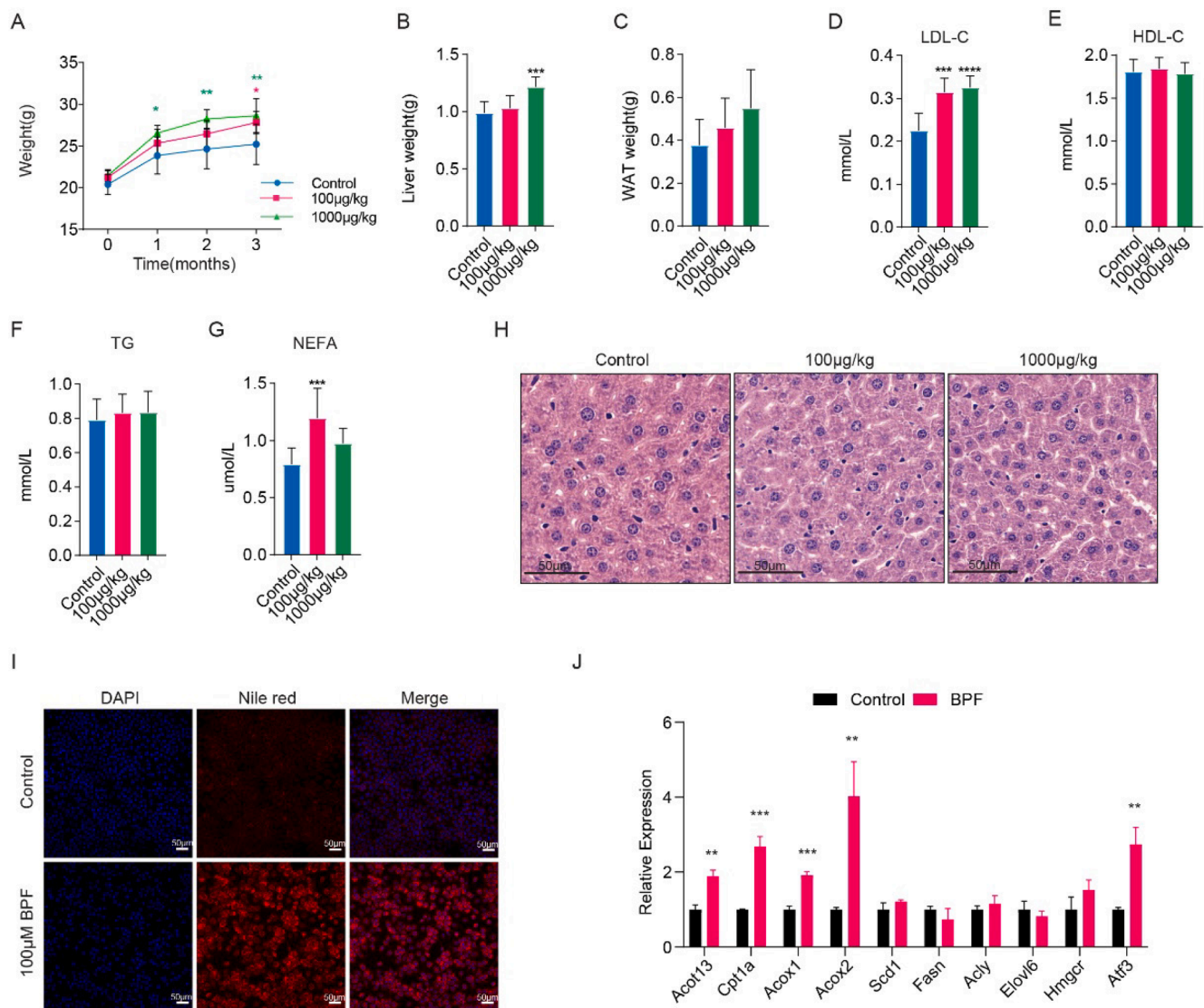


Fig. 1. BPF promotes lipid deposition in hepatocytes. (A) Body weight (mean $\pm$ SD) of study animals ($n = 8$ per group) in control, BPF-L (100 μ g/kg bw/day) and BPF-H (1000 μ g/kg bw/day) group. (B–C) Liver and white fat tissues weight in control, BPF-L (100 μ g/kg bw/day) and BPF-H (1000 μ g/kg bw/day) group ($n = 7–8$ per group). (D–G) Serum low-density lipoprotein cholesterol (LDL-C), high-density lipoprotein cholesterol (HDL-C), triglyceride (TG) and non-esterified fatty acids (NEFA) in control, BPF-L and BPF-H group ($n = 7–8$ per group). (H) Representative hematoxylin and eosin (H&E) staining of liver in control, BPF-L and BPF-H group. (I) Nile Red staining of AML12 cells in control and BPF (100 μ M) group. (J) Relative mRNA expression levels of markers of lipid metabolism in AML12 cells. Significance was calculated using non-paired two-tailed Student's t test or one-way ANOVA analysis. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, **** $P < 0.0001$. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

3.2. BPF exposure reprogrammed the hepatic transcriptome

To further evaluate the hepatic transcriptome changes after BPF exposure, we performed RNA-seq in mouse liver tissue (BPF vs Control: 1000 μ g/kg bw/day vs 0 μ g/kg bw/day) and AML12 cell line (BPF vs Control: 100 μ M vs 0 μ M). *In vivo* exposure models, 3388 DEGs were identified after BPF exposure ($FDR < 0.05$, $|\log_2\text{FoldChange}| \geq 1$), comprising 1543 significantly upregulated DEGs and 1845 down-regulated DEGs (Fig. 2A, S2A, Supplementary Table 3). As shown in Fig. 2B, GO analysis showed that these DEGs were enriched in process related to the regulation of small molecule metabolic, generation of precursor metabolites and energy, ATP metabolic, response to endoplasmic reticulum stress pathways. Furthermore, the KEGG pathway analysis revealed that protein processing in endoplasmic reticulum, oxidative phosphorylation, non-alcoholic fatty liver disease pathways were enriched (Fig. 2C). Using DEG analysis, we found 94 genes were upregulated and 174 genes downregulated ($N = 31,058$, $FDR < 0.05$, |

$\log_2\text{FoldChange}| \geq 1$; Fig. 2D, S2B, Supplementary Table 4) following BPF exposure. According to GO analysis, DEGs were enriched in metabolisms related to sulfur compound metabolism, steroid metabolism, and carboxylic acid biosynthetic process (Fig. 2E). Moreover, gene set enrichment analysis (GSEA) showed that ERS pathway was significant enriched in BPF exposure group (Normalized enrichment score, NES = 1.650, $P = 0.003$; Fig. 2F). In BPF exposure group, we observed alteration in lipid metabolism was characterized by increasing of genes involved in *de novo* lipogenesis (DNL) (*Scd1*, *Ces1d*, *Srebf2*, *Plin5*, *Acsl4*, *Acsl5*) (Fig. 2G), which was crucial for the storage of energy as triglycerides in hepatocytes. Intriguingly, BPF exposure altered hallmark genes associated with ERS and URP with the upregulation of *Atf6*, *Ern1*, *Eif2a* and the downregulation of *Ddit3*, *Atf4*, *Cebpb* (Fig. 2H–I). Collectively, these findings suggest that BPF exposure reprograms ERS and lipid metabolism related gene networks linked to NAFLD.

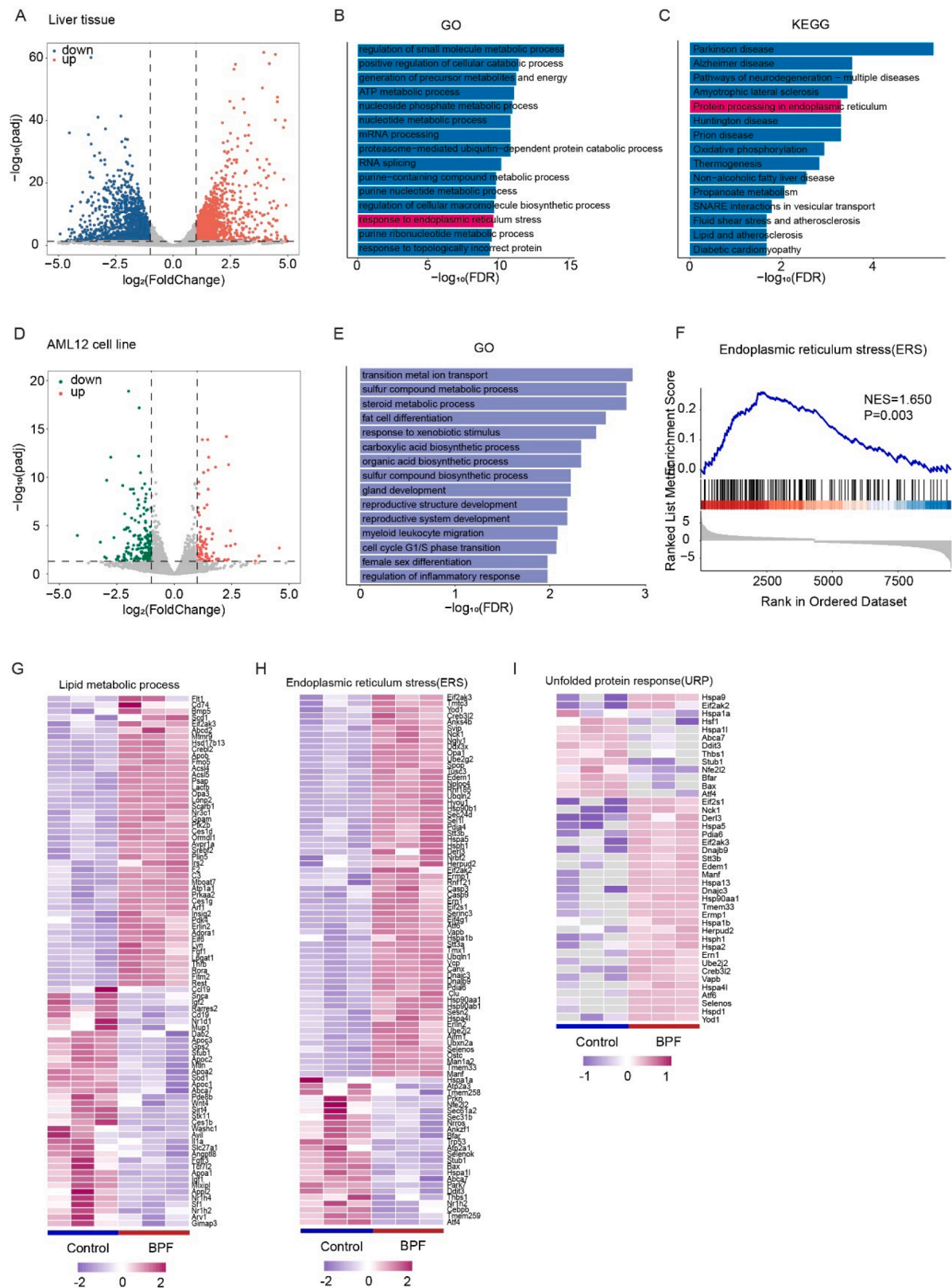


Fig. 2. BPF reprogrammed the transcriptomics in liver and AML12 cell. (A) Hepatic volcano plot of BPF altered differentially expressed genes in mice (BPF: 1000 $\mu\text{g/kg}$ bw/day, $n = 3$ per group). (B-C) GO and KEGG pathway analysis of differentially expressed genes in mice. (D) Volcano plot of BPF altered differentially expressed genes in AML12 cells (BPF: 100 μM , $n = 3$ per group). (E) GO analysis of differentially expressed genes in AML12 cells. (F) Gene set enrichment analysis (GSEA) analysis of differentially expressed genes in endoplasmic reticulum stress pathway. (G-I) Hepatic heatmap of key genes involved in lipid metabolic process, endoplasmic reticulum stress (ERS) and unfolded protein response (URP) ($n = 3$ per group). Each column contains data from an individual study sample and each row represents a gene. Significance was calculated using non-paired two-tailed Student's t test. $*P < 0.05$, $**P < 0.01$, $***P < 0.001$, $****P < 0.0001$. Multiple testing correction was calculated. $*FDR < 0.05$ was determined statistically significant.

3.3. BPF exposure changed hepatic metabolic profiles

To figure out how BPF disturbs the hepatic profile of metabolites, liquid chromatograph mass spectrometer (LC-MS/MS) untargeted metabolomics was performed to detect metabolites of liver tissues in the control and BPF groups (1000 µg/kg bw/day, 8 samples per group). PCA and OPLS-DA indicated significant differences between the BPF-treated and control groups (Fig. S3A-B). We determined 110 differential expression metabolites (DEMs) were increased and 61 DEMs were decreased ($N = 171$, $VIP > 0$, P -value < 0.05 , $|\log_2\text{FoldChange}| \geq 1$; Fig. 3A-B, Supplementary Table 5). Metabolic enrichment pathway analysis showed purine metabolism, ether lipid metabolism, pantothenate and CoA biosynthesis were altered after BPF exposure (Fig. 3C).

3.4. Joint analysis of the hepatic transcriptome and metabolome characterized the hepatotoxicity role of ERS

To elucidate the underlying hepatic adaptation pattern when respond to exogenous stress, we used MetaboAnalyst joint pathway analysis to investigate the effects of BPF exposure on the network of hepatic metabolome and transcriptome. The result revealed that DEGs and DEMs were significantly enriched in PI3K-Akt signaling pathway, thermogenesis, non-alcoholic fatty liver disease (NAFLD), and protein processing in endoplasmic reticulum (Fig. 4A). Consistent with our previous results, DEM-DEG regulatory network showed the protein processing in endoplasmic reticulum pathway with dysregulated expression of genes (Fig. 4B), which suggested BPF exposure disrupted the hepatic lipid metabolism via ERS and its related genes.

3.5. BPF modified the chromatin accessibility in AML12

Gene expression is coordinated by regulatory elements and chromatin accessibility, which characterizes the degree of chromatin openness providing for TFs binding. To identify the chromatin accessibility altered by BPF, we performed ATAC-seq in AML12 cells. In total, BPF exposure induced 2533 loci with significantly differential enriched signal by EdgeR analysis ($FDR < 0.05$, $\text{Foldchange} > 2$, Fig. 5A-B, Supplementary Table 6). Of which, 1075 exhibited increased accessibility and 1458 exhibited decreased accessibility. Enhancers are usually defined as the peak regions of enhancer markers (such as H3K27ac and H3K4me1), which are located at least 5 kb away from the annotated TSS (Qin et al., 2022; Rada-Iglesias et al., 2011). Remarkably, both BPF gain and loss regions were distanced from transcription start site (TSS) regions (Fig. 5C). Next, the location of BPF gain and loss ATAC peaks was determined by ChIPseeker via genomic characteristics classified as the intergenic, exon, promoter, and intro. In both BPF gain and loss regions, more than 25 % peaks were located far from the gene promoters (Fig. 5D), indicating that these regions may function as enhancers. In line with previous findings from the metabolome and transcriptome pathway results, the GO analysis showed that these differentially accessible regions were enriched with genes linked with metabolic and ERS pathways, including steroid metabolic process, cellular response to insulin stimulus, and negative regulation of protein serine/threonine kinase activity (Fig. 5E). Example loci depicting differentially accessibility and expression were visualized in Fig. 5F.

TFs found in enhancer regions are typically bind cognate *cis*-acting DNA sequences to control gene expression. To explore differential TFs occupancy at differentially accessible regions after BPF exposure, we

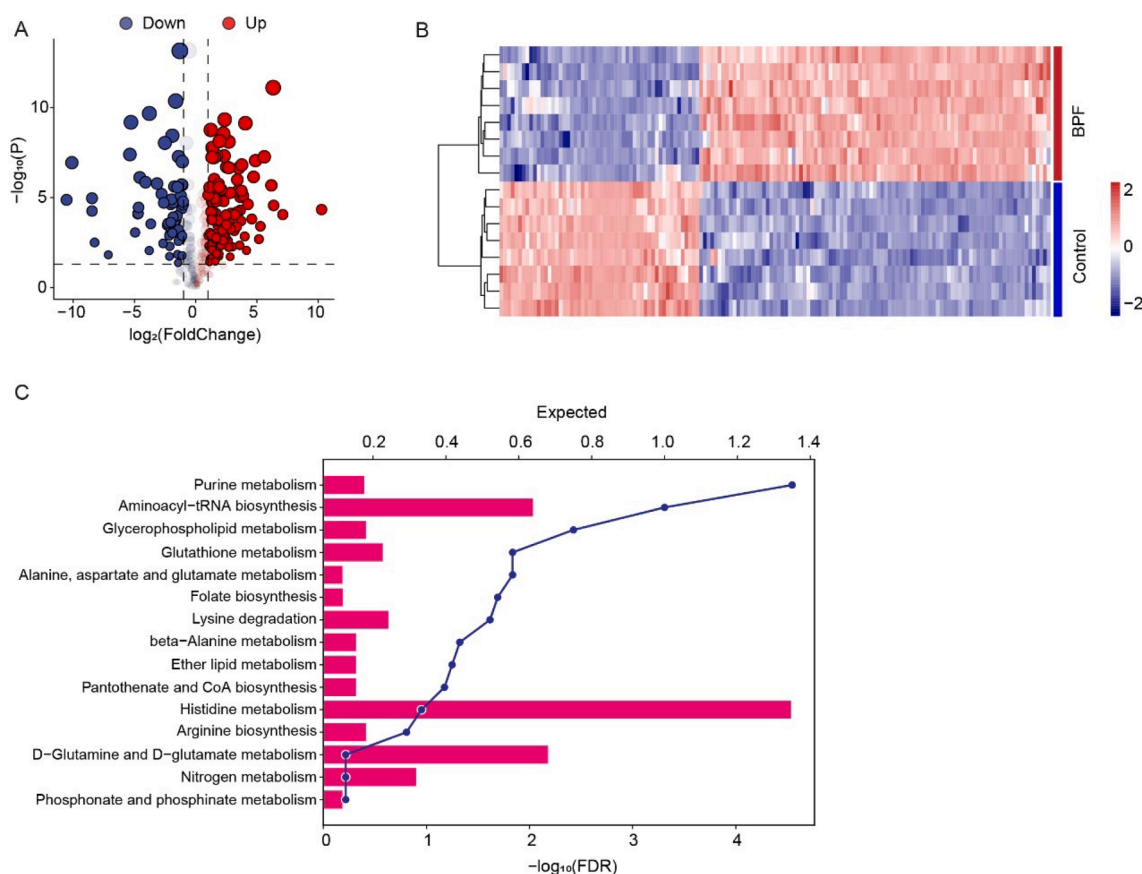


Fig. 3. BPF modulates metabolic profile in liver. (A-B) Volcano plots and heatmap of differentially expression metabolites between control and BPF groups (BPF: 1000 µg/kg bw/day, $n = 8$ per group). (C) KEGG pathway analysis of differentially expressed metabolites. Significance was calculated using non-paired two-tailed Student's *t* test. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, **** $P < 0.0001$. Multiple testing correction was calculated. * $FDR < 0.05$ was determined statistically significant.

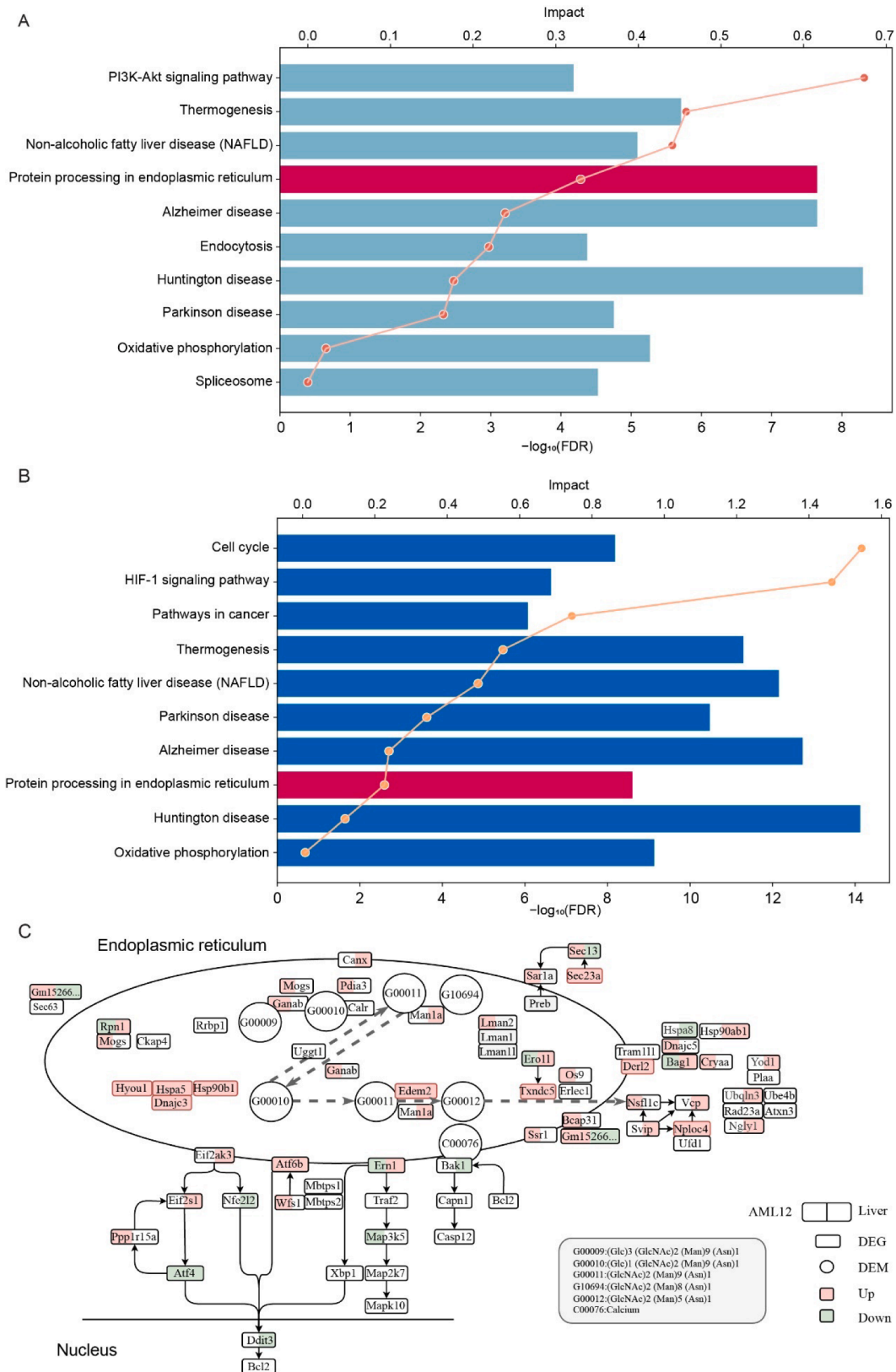


Fig. 4. Integrated analysis of transcriptome and metabolome. (A-B) KEGG pathway analysis of differentially expressed genes and differentially expressed metabolites in mice and AML12 cells. (C) Schematic representation of endoplasmic reticulum stress pathways associated with differentially enriched metabolites and genes. Metabolites/genes increased in BPF group are indicated by red filled modules and metabolites/genes decreased in BPF group are marked by green filled modules. Significance was calculated using non-paired two-tailed Student's *t* test. **P* < 0.05, ***P* < 0.01, ****P* < 0.001, *****P* < 0.0001. Multiple testing correction was calculated. **FDR* < 0.05 was determined statistically significant. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

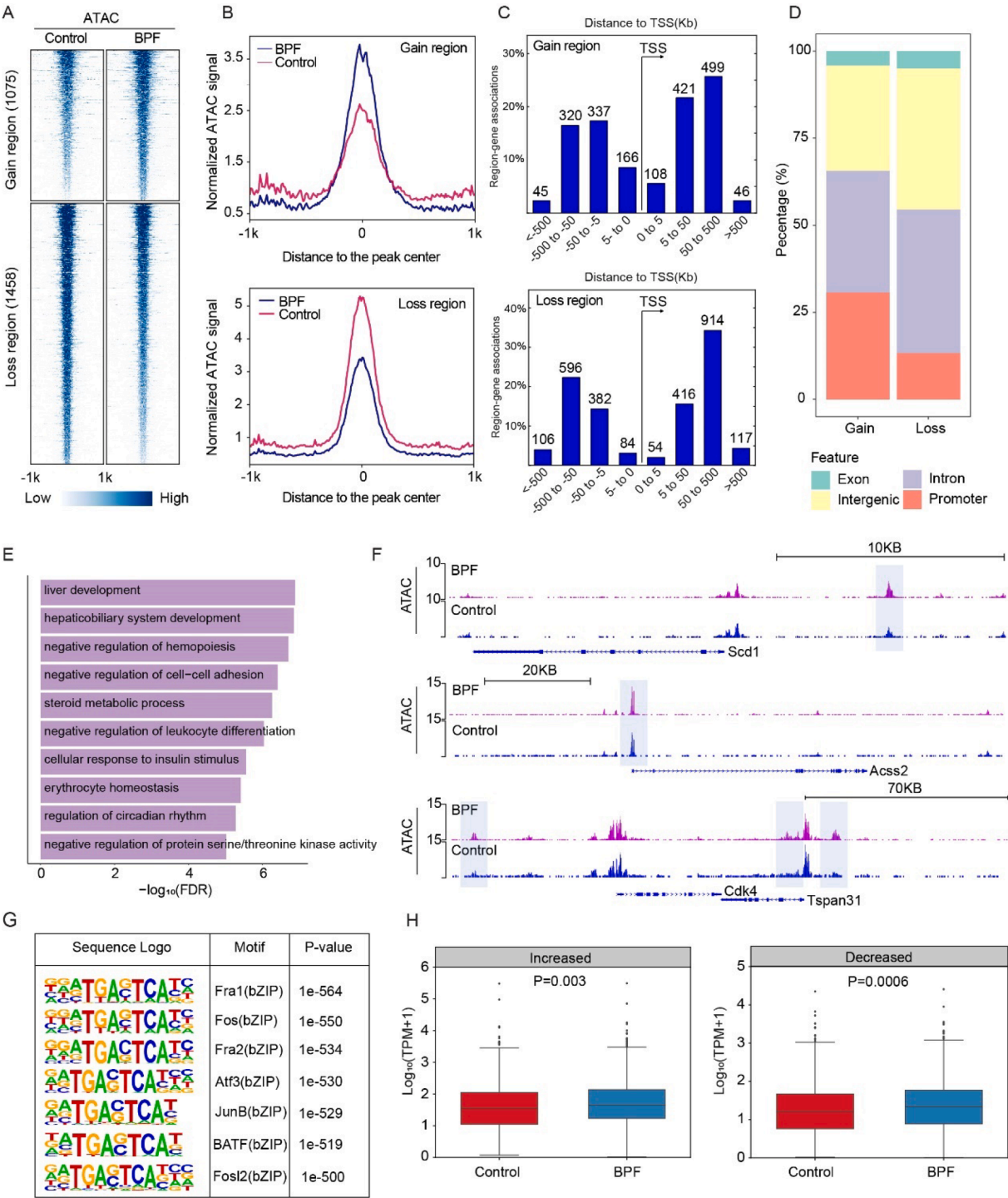


Fig. 5. BPF modifies the chromatin accessibility. (A) Heatmap of BPF induced differential chromatin accessible regions in control and BPF group (n = 2 per group). (B) Metagenes plot of ATAC signal at differential chromatin accessible regions. (C-D) Annotation of differential chromatin accessible regions. (E) GO pathway analysis of loci near differential chromatin accessible regions. (F) Examples of differential chromatin accessible regions near gene promoters. (G) Motifs enriched at differential chromatin accessible regions. (H) Gene expressions in regions with increased or decreased accessibility. Significance was calculated using non-paired two-tailed Student's *t* test. **P* < 0.05, ***P* < 0.01, ****P* < 0.001, *****P* < 0.0001. Multiple testing correction was calculated. *FDR < 0.05 was determined statistically significant.

performed HOMER *de novo* motif algorithm analysis. Our analysis found that a basic-region leucine zipper (bZIP: Fra1, Fos, Fra2, Atf3, JunB, BATF and Fosl2) family transcription factor with the binding site characterized “TGA(C/G)TCA” across differential accessible regions (Fig. 5G). The ATF/CREB members share the bZIP domain that compose a leucine zipper dimerization motif and a DNA-binding basic region.

ATF/CREB is a conserved protein family can function as transcription factors or cofactors to regulate lipid homeostasis in metabolic tissues, especially in liver. Next, we asked whether BPF-altered chromatin accessibility regulate the change of gene expression. the analysis revealed that increased chromatin accessibility was significantly associated with higher genes expression (Fig. 5H).

3.6. BPF changed the dynamic enhancer landscape in AML12

As histone H3 acetylation at lysine 27 (H3K27ac) is a well-established chromatin marker of active enhancers and promoters that relate to increased chromatin accessibility, transcription factor binding and gene transcription. We sought CUT&Tag to delineate the histone modifications in AML12 cells with or without BPF exposure. We

observed about 9387 loci displaying significantly differential H3K27ac enrichment due to BPF exposure (3392 loss regions and 5995 gain regions; $FDR < 0.05$, Foldchange > 2 , Fig. 6A-B). Interestingly, BPF gain regions were distant from TSS regions, while BPF loss regions were adjacent with TSS regions (Fig. 6C). Through ChIPseeker, we assigned peaks to genomic annotation in the intergenic, exons, promoters, and introns. We observed less than 50 % peaks were located at the gene

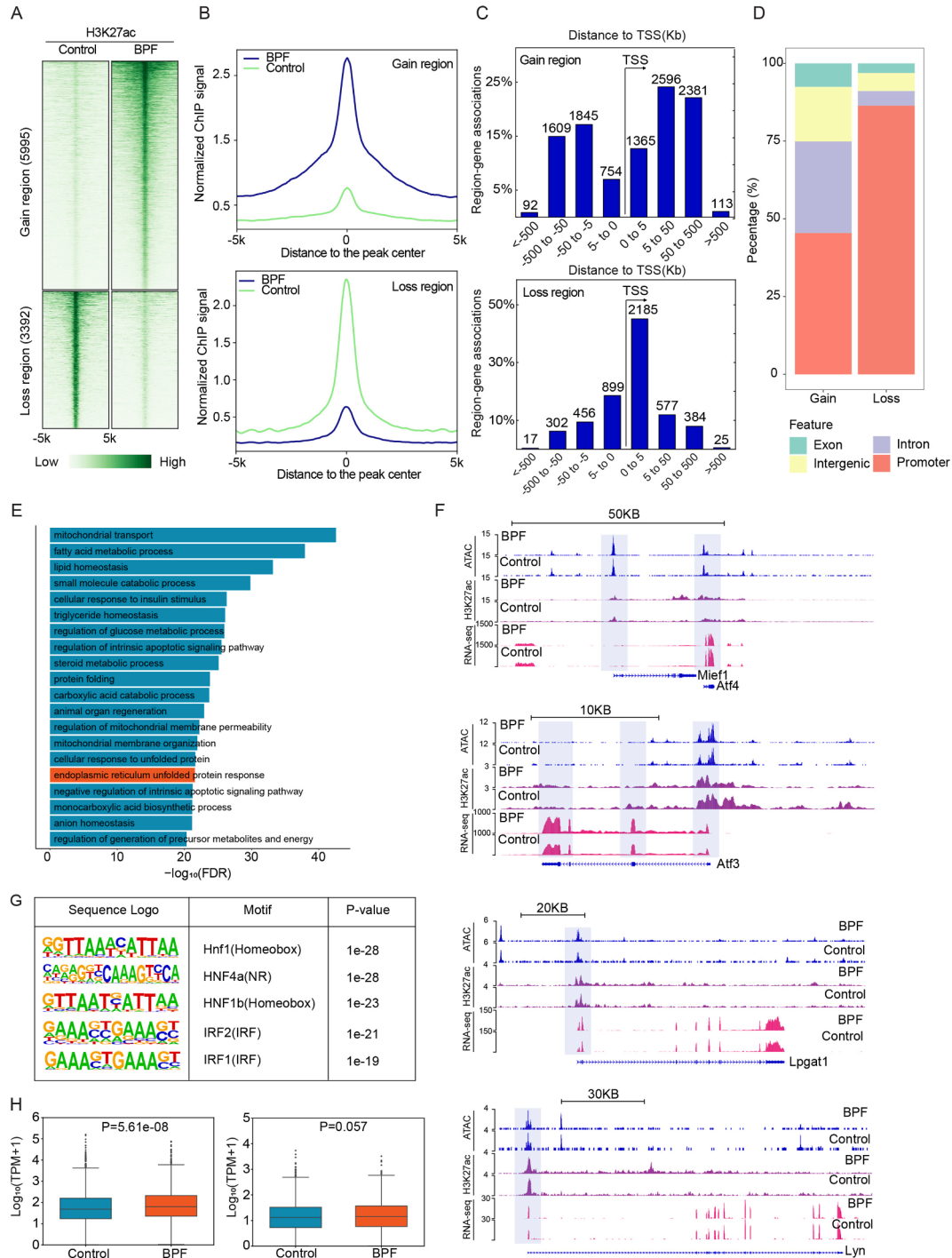


Fig. 6. BPF alters the active enhancer landscape. (A) Heatmap of H3K27ac signal in control and BPF group (n = 2 per group). (B) Metagenes plot of H3K27ac signal in control and BPF group. (C-D) Annotation of differential H3K27ac enriched regions. (E) GO pathway analysis of loci near differential H3K27ac enriched regions. (F) Examples of differentially expressed genes with differential chromatin accessible regions and H3K27ac enriched regions. (G) Motifs enriched at differential H3K27ac enriched regions. (H) Gene expressions in regions with increased or decreased H3K27ac enrichment. Significance was calculated using non-paired two-tailed Student's *t* test. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, **** $P < 0.0001$. Multiple testing correction was calculated. * $FDR < 0.05$ was determined statistically significant.

promoters in BPF gain regions. However, there were about 90 % peaks resided in gene promoters in BPF loss regions (Fig. 6D).

To decipher the underlying biological pathway with these differential binding regions, we used GO analysis. As anticipated, we found that those regions were enriched with genes linked to pathways comprising fatty acid metabolic process, lipid homeostasis, triglyceride homeostasis, endoplasmic reticulum unfolded protein response (Fig. 6E). Example loci depicting differentially accessibility and expression were presented in Fig. 6F. To identify potential TFs, we used HOMER motif analysis and found that the top enriched motifs were related to Homebox family's known consensus binding sequences (Hnf1 and Hnf1b), nuclear receptor (NR) family (Hnf4a) and interferon regulatory factor (IRF) family (IRF1 and IRF2) (Fig. 6G). Increased H3K27ac signals showed significant correlation with higher gene expressions (Fig. 6H). These data suggest that BPF reprograms enhancer landscape integrating to ERS.

4. Discussion

As a widely used BPA substitute, concerns are growing regarding the health effect of BPF exposure. In this study, we found that BPF exposure disrupted hepatic lipid metabolism *in vivo* and *in vitro*. Through the integration of RNA-seq, ATAC-seq, and CUT&Tag, we revealed that BPF exposure altered hepatic chromatin accessibility and enhancer landscape. Interestingly, we found that enrichment of bZIP family TFs within open chromatin regions induced by BPF. Relocation of these bZIP TFs might lead to the reprogramming of downstream gene sets such as ERS and lipid metabolic genes. Overall, this study uncovered a critical role of ERS in BPF hepatic toxicity, expanding our understanding of the epigenetic mechanisms in NAFLD.

Growing evidence demonstrated that BPF exposure induced hepatic steatosis, including increased triacylglycerol and cholesterol level, elevated lipid droplets *in vivo* and *in vitro* (Meng et al., 2018; Wang et al., 2021; Zhang et al., 2023). As expected, we found exposure to BPF could result in hepatic steatosis both *in vivo* and *in vitro*. Although bisphenol A (BPA) and its alternatives exposure disrupted hepatic lipid and glucose homeostasis to varying degrees, BPF exposure was more pronounced than that induced by BPA (Sun et al., 2023).

To fill in the gap of the underlying mechanism, we integrated multi-omics approach including RNA-seq, metabolome, ATAC-seq and CUT&Tag in livers and AML12 cells with or without BPF treatment. In lines with our study, previous study also suggested that BPF exposure increased expression of hepatic stearoyl-CoA desaturase 1 (*Scd1*), which was a key molecule in the regulation of hepatic lipogenesis and lipid oxidation (Mauvoisin et al., 2007). Moreover, DEGs were discovered to be enriched in the steroid biosynthesis and lipid metabolism pathways (Sun et al., 2023). In our study, we observed that ERS pathway were significant enriched in joint analysis of the hepatic transcriptome and metabolome, suggesting that BPF exposure might disrupt the hepatic lipid metabolism via ERS. It is well documented that ERS promoted NAFLD and other multiple hepatic pathologies (Ajoalabady et al., 2023; Lebeaupin et al., 2018).

Liver is the major metabolic organ and endoplasmic reticulum is the major organelle for lipid synthesis in hepatocytes. ERS is an adaptive response in cells triggered by the aggregation of unfolded proteins or misfolded proteins in the ER, which impairs the normal physiological functions of the endoplasmic reticulum. Upon activation of the unfolded protein response, three ER membrane sensors, activating transcription factor 6a (ATF6a), inositol requiring (IRE) 1a, and PKR-like ER-localized kinase (PERK), mediating signals from the endoplasmic reticulum (Ajoalabady et al., 2023). Numerous disturbances such as inflammation, viruses, drugs, hyperlipidemia, were proved to impair hepatocyte ER homeostasis, facilitating the dysregulation of hepatic lipid metabolism and liver disease (Liu et al., 2021; Song et al., 2021). In our study, we found BPF exposure, as a disturbance, reprogrammed the gene sets of endoplasmic reticulum stress and unfolded protein response including upregulation of *Atf6*, *Ern1*, *Eif2a* and downregulation of *Ddit3*, *Atf4*,

Cebpb. In addition, sterol regulatory element-binding proteins govern *de novo* lipogenesis, with SREBP1c (*Srebp1*) is engaged in fatty acid synthesis and SREBP2 (*Srebp2*) involved in sterol production. Consistently, our data revealed that BPF exposure altered sterol synthesis metabolism pathway that related to the development of NAFLD (Kang et al., 2022; Ren et al., 2023).

A growing body of evidence indicates that epigenetic regulation, including chromatin accessibility, histone modifications, various non-coding RNAs and DNA methylation, played key roles in regulating gene expression (Fan et al., 2023; Perino and Veenstra, 2016; Slack and Chinnaiyan, 2019). Remodeling chromatin accessibility can simultaneously reprogram multiple gene sets and various signaling pathways in disease progression (Perino and Veenstra, 2016). In our study, we demonstrated that BPF exposure modulated chromatin state, with a majority of differential regions corresponding to potential enhancers. As expected, increased chromatin accessibility was significantly associated with higher gene expression relying on TFs regulatory networks and a series of chromatin modifiers. Motif analysis showed that a bZIP TF family including Fra1, Fos, Fra2, Atf3, JunB, BATF, and Fosl2 was occupied depended on the chromatin accessibility under BPF exposure. Among those, ATF3 is a stress-responsive transcription factor belonging to ATF/CREB protein family. As a conserved protein family, the ATF/CREB family members (ATF1-ATF7) owned with a basic-region leucine zipper (bZIP) domain in common, and deregulated ATF/CREB family signaling is implicated in the development of NAFLD, type 2 diabetes, and cancer (Cui et al., 2021). In our model, we found BPF exposure changed the expressions of Atf3, Atf4 and Atf6 and they were all involved in ERS. These ATF/CREB proteins could be activated by ERS signaling cascade and participated in lipogenesis process or fatty acid oxidation via regulating the downstream target genes such as *Srebf1*, *Pparg*, *Fgf21*, *Sphk2*, and *Srebf2* (Okada et al., 2003; Cui et al., 2021).

5. Conclusion

Our current study expands knowledge about how chromatin accessibility and transcriptional network respond to environmental cues. Though multi-omics approach, we have uncovered that BPF exposure modifies the chromatin landscape and bZIP TFs binding to regulate genes involved in ERS and lipid metabolic gene networks, which contribute to NAFLD. Therefore, our findings highlight a novel epigenetic mechanistic connection between BPF exposure and fatty liver.

CRedit authorship contribution statement

Yun Fan: Writing – original draft, Visualization, Funding acquisition, Formal analysis, Conceptualization. **Shiqi Li:** Validation, Methodology, Investigation, Data curation. **Xiancheng Yang:** Validation, Methodology, Investigation. **Shengjun Bai:** Visualization, Software, Formal analysis, Data curation. **Min Tang:** Resources, Methodology, Data curation. **Xueer Zhang:** Visualization, Methodology, Formal analysis. **Chuncheng Lu:** Validation, Supervision, Project administration. **Chenbo Ji:** Validation, Supervision. **Guizhen Du:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Yufeng Qin:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.envint.2024.108690>.

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